Oscillations

Goals For Today

- Motivate why we care about simple harmonic motion (SHM)
- Solve the equation for SHM
- Modify SHM so that we can use the same ideas in more situations

What happens when we are at the minima of a potential?

Say we have a system that has some potential $V(x)$, and we put a particle at a minima of that potential. We then poke the particle some small amount Δx away from the minima. We want to know what happens.

1. Why might we care about this situation?

2. How can we approximate what the potential looks like near the minima? Make the approximation. *Hints: how big is our movement? What approximation works well for small changes? If we are at a minima, what is true about the first derivative of $V(x)$? Feel free to define a constant parameter as necessary.*

3. What is the relationship between the force and potential? Use this to set up an equation of motion for the particle. What do you notice? Does it look familiar?

Solving simple harmonic motion

You should have gotten the differential equation

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x \tag{1}$$

(You could also have written it as $\ddot{x} + \frac{k}{m}x = 0$.)

This is arguably the most important differential equation in all of physics, and one that a lot of professors solve by 'guessing,' by which they really mean 'I have seen this so many times I know what the answer should look like.' Instead, let's solve this using the technique we learned earlier: understanding function behavior.

1. What is this differential equation *saying* in words? What kind(s) of function(s) might solve it?

2. Use your previous answer to write a general solution to the differential equation. *Hint: you should have gotten at least two answers to the first question. How can we combine two types of solutions to make one general solution? Introduce constants as necessary, but be sure to explain what these constants represent.*

3. What kind of motion is this physically describing? What systems can you think of that have this sort of motion? (Don't limit yourself to what you've covered so far in Physics 5A/7A - there are lots you may have encountered in day to day life!)

4. Spoiler alert: you probably wrote an answer down in terms of two different trigonometric functions. Use the identity $A\cos(x) + B\sin(x) = C\cos(x+\phi)$, with $C = \text{sign}(A)\sqrt{A^2+B^2}$ and $\phi = \arctan(-B/A)$ to write your result in the form

$$x(t) = A\cos\left(\sqrt{\frac{k}{m}}t + \phi\right) \tag{2}$$

Use this result with your answer to question two of the first section of this worksheet to determine the energy in the system. Does your result make sense? What kind of quantity must A be, and what physical meaning does it have in the system? *Hint: don't forget about kinetic energy!*

5. Trigonometric functions, however, kind of suck.[1] What sort of function is really nice to work with? How could we write our solution from the previous part in terms of that function?[2] *Hint: Euler's identity is a thing of beauty and a joy forever.*[3] *You are allowed to take only the real or imaginary component of a result. Alternatively, return to question one, and see how you could modify the nice function you picked to make it 'fit' the differential equation.*

6. List all of the different formats we've seen so far that oscillations/SHM can take. It's important to be able to recognize it when it shows up! You should be able to identify at least six.

[1] [Citation needed.]

[2] This is one of the best techniques for solving differential equations in physics: plug in this function with a few constant parameters you can adjust to your differential equation, and change the parameters so it works.

[3] Memorize it if you haven't yet, after proving it's true using Taylor series.

7. It's important to understand all of the parameters a system has. What units does $\sqrt{\frac{k}{m}}$ have? What kind of quantity must it be? Because of this, we often write $\omega_0^2 = \frac{k}{m}$, and call ω_0 the *resonant frequency*, for reasons we'll see starting next week.

Examples

1. **Hole through the Earth.** Imagine you drill a hole through the Earth, and jump in. Assume the Earth has a constant density. Will you return to your starting location? If so, how long will it take?

   (a) Visualize: what will your motion look like? Imagine jumping into this hole yourself! (Neglecting any pressure or temperature issues, of course.)

   (b) Use dimensional analysis to see what form your result might take.

   (c) What other problems might be similar to this?

   (d) Set up an equation of motion. Hint: use the relationship between mass, volume, and density.

   (e) Solve. Does your result make sense?

(f) Find the time it will take to return to your starting location. Does your result make sense?

2. A particle of mass m is free to move in the region $0 \leq x \leq \infty$. It is governed by the potential energy

$$U(x) = U_0\Big(\frac{x}{A} + \lambda^2\frac{A}{x}\Big)$$

Where U_0, A, and λ are constants.

(a) Sketch roughly what this potential looks like.

(b) Find the equilibrium distance x_0, where the particle is subject to no force.

(c) Let $\Delta x = x - x_0$ be the distance of the particle from equilibrium. Under conservation of energy, what is the frequency of simple harmonic oscillations about equilibrium when Δx is very small?

3. **Oceans, atmospheres, and convection.** In a fluid (like air or water), convection occurs - that is, circular vertical currents. Consider a small chunk of fluid particles with density ρ_0, and with the density a function of height[4] ($\rho = \rho(z)$). If the fluid chunk is displaced vertically by some small height Δz (and maintains the same density ρ_0 within the parcel), it will experience a change in the gravitational force.

(a) Draw a diagram of the situation. Then draw a free body diagram for the parcel before being moved up a height Δz and after being moved up a height Δz.

[4]Check that why this is so makes sense to you! Is the air density the same at the top of Mount Everest as it is at sea level?

(b) Write the change in the gravitational force on the parcel.

(c) Use the previous part and $F = ma$ to set up a differential equation for the displacement Δz of the parcel.

(d) Ugh, this looks gross. Let's linearize! We won't even formally Taylor expand here. What looks gross in this equation? What quantity is small? How can we approximate it? (Hint: what is the definition of a derivative?)

(e) Solve this linearized differential equation.

(f) What happens when the density decreases with height? When it increases? What does this tell you about what the fluid is doing in each case? This frequency is called the *Brunt–Väisälä frequency*.

Reflection

Take some time to think through these questions. Part of purposeful practice is thinking through what your practice has taught you, how you will use it in the future, and what you might do differently next time, so we want to give you the space to do this.

1. What did you learn from these problems? How did you know to use the techniques you used? How might you apply this to a future problem?

2. How might you connect this to past material you've learned?

3. What questions do you have?